

Lecture-2. Python fundamentals.

Python While Loops

What is a While Loop?

A while loop allows a block of code to be executed repeatedly as long as a given condition is true. It is useful when the number of iterations is not known in advance.

Syntax:

```
while condition:  
    # Code block
```

Example:

```
count = 0  
while count < 5:  
    print("Count:", count)  
    count += 1
```

Output:

```
Count: 0  
Count: 1  
Count: 2  
Count: 3  
Count: 4
```

```
import random

number_to_guess = random.randint(1, 10)
guess = 0

while guess != number_to_guess:
    guess = int(input("Guess the number (1-10): "))
    if guess < number_to_guess:
        print("Too low!")
    elif guess > number_to_guess:
        print("Too high!")
    else:
        print("Congratulations! You guessed it!")
```

Python Lists

What is a List?

A list is a collection of ordered, mutable items in Python. Lists allow duplicate elements and can store mixed data types.

Syntax:

```
my_list = [item1, item2, item3]
```

Example:

```
fruits = ["apple", "banana", "cherry"]
fruits.append("orange")
print(fruits)
```

Output:

```
['apple', 'banana', 'cherry', 'orange']
```

Indexing in lists:

Index	0	1	2	3
Value	a	b	c	d

List Operations:

1. **Adding Items:** `list.append()`, `list.insert()`
2. **Removing Items:** `list.remove()`, `list.pop()`

`list.append()` :

Adds an item to the end of the list.

```
my_list = [1, 2, 3]
my_list.append(4)
print(my_list) # Output: [1, 2, 3, 4]
```

`list.insert()` :

Inserts an item at a specified index.

```
my_list = [1, 2, 3]
my_list.insert(1, 10) # Inserts 10 at index 1
print(my_list) # Output: [1, 10, 2, 3]
```

`list.remove()` :

Removes the first occurrence of a specified item from the list. If the item is not found, it raises a `ValueError`.

```
my_list = [1, 2, 3, 2]
my_list.remove(2) # Removes the first occurrence of 2
```

```
print(my_list) # Output: [1, 3, 2]
```

list.pop() :

Removes an item at a specified index and returns it. If no index is specified, it removes and returns the last item. If the list is empty, it raises an `IndexError`.

```
my_list = [1, 2, 3]
removed_item = my_list.pop(1) # Removes and returns the item
                               # at index 1
print(removed_item) # Output: 2
print(my_list) # Output: [1, 3]
```

If you don't provide an index:

```
my_list = [1, 2, 3]
removed_item = my_list.pop() # Removes and returns the last
                               # item
print(removed_item) # Output: 3
print(my_list) # Output: [1, 2]
```

Python For Loops

What is a For Loop?

A for loop iterates over a sequence (like a list, tuple, or range) and executes the block of code for each item.

Syntax:

```
for variable in sequence:
    # Code block
```

Example:

```
for i in range(5):  
    print("Iteration:", i)
```

Output:

```
Iteration: 0  
Iteration: 1  
Iteration: 2  
Iteration: 3  
Iteration: 4
```

Python Arrays

What is an Array?

Arrays are a collection of items stored at contiguous memory locations. In Python, the `array` module can be used to work with arrays.

Syntax:

```
import array  
arr = array.array('i', [1, 2, 3])
```

Example:

```
import array  
a = array.array('i', [1, 2, 3, 4])  
a.append(5)  
print(a)
```

Output:

```
array('i', [1, 2, 3, 4, 5])
```

Python Tuples

What is a Tuple?

A tuple is an immutable sequence of items. Tuples are ordered and can contain duplicate elements.

Syntax:

```
my_tuple = (item1, item2, item3)
```

Example:

```
fruits = ("apple", "banana", "cherry")  
print(fruits[1])
```

Output:

```
banana
```

Tuple Characteristics:

1. Immutable.
2. Allow duplicate elements.

Python Sets

What is a Set?

A set is an unordered collection of unique items. Sets are mutable and do not allow duplicate values.

Syntax:

```
my_set = {item1, item2, item3}
```

Example:

```
unique_numbers = {1, 2, 3, 3, 4}
print(unique_numbers)
```

Output:

```
{1, 2, 3, 4}
```

Python Dictionaries

What is a Dictionary?

A dictionary is an unordered collection of key-value pairs. Keys must be unique and immutable.

Syntax:

```
my_dict = {key1: value1, key2: value2}
```

Example:

```
student = {"name": "Alice", "age": 25}
print(student["name"])
```

Output:

```
Alice
```

Operations:

1. **Access Value:** `dict[key]`
2. **Add/Update Key-Value Pair:** `dict[key] = value`
3. **Delete Key-Value Pair:** `del dict[key]`